

LDNA Expressiveness

1 Expressiveness of LDNA

This section characterises the representational power of the Learnable Dynamic Neighborhood Aggregation model (LDNA). The reference point is the one-dimensional Weisfeiler–Leman test (1-WL) of Weisfeiler and Leman [1968], the standard yardstick for message-passing graph neural networks [Xu et al., 2019, Morris et al., 2019]. All statements are about the model *class*: they concern the existence of parameters realising a given behaviour, not about any particular trained instance. Throughout, $\{\{\cdot\}\}$ denotes a multiset (a set with repetitions), and for a finite multiset M over a set A we write $M(a)$ for the multiplicity of $a \in A$.

1.1 Setup

Attributed graphs. An attributed graph is $G = (V, E, x, e)$ with node set V , directed edge set $E \subseteq V \times V$ (an undirected edge is the pair of its two orientations), a node-feature map $x : V \rightarrow \Sigma$ and, when present, an edge-feature map $e : E \rightarrow \Xi$, $e_{uv} := e(u, v)$. We write $N(v) = \{u : (u, v) \in E\}$ for the in-neighbourhood and $n = |V|$. We make the following standing assumptions.

- **Finite alphabets, bounded size.** The node alphabet $\Sigma \subset \mathbb{R}^{d_0}$ and the edge alphabet Ξ are finite, and graph sizes are bounded, $n \leq N$. This is what makes exact, finite-domain constructions available: an MLP with ReLU activations can interpolate *any* prescribed function on a finite input set exactly (Lemma 1), so no appeal to universal approximation is needed anywhere below.
- **Minimum degree one.** Every node has at least one neighbour, $N(v) \neq \emptyset$. Unlike the previous item this is not universally satisfied by the datasets — isolated nodes do occur in practice (e.g. single-atom fragments in molecular graphs) — so we invoke it only where it is stated. It is used solely by the lower-bound constructions (Proposition 1, the $\succeq$ direction of Proposition 2, Theorem 1(a) and Theorem 2), to let a node’s own colour be recovered from its non-empty aggregation; the upper-bound statements (Theorem 1(b), Proposition 3 and the $\preceq$ direction of Proposition 2) hold for all graphs without it. Its necessity is examined in Remark 2.

The LDNA layer. Nodes carry hidden states $h_v^{(t)} \in \mathbb{R}^{d_t}$, and each node has a scalar *canonical rank* $r(v) \in [0, 1]$ fixed once per graph (defined below). One LDNA layer (the module **LDNAConv**) computes, for every directed edge $u \rightarrow v$ with $u \in N(v)$, a pre-message, a per-channel gate of the rank gap, and a post-transform of the gated sum:

$$m_{uv} = \text{pre}([h_v^{(t)}, h_u^{(t)}, e_{uv}]), \quad (1)$$

$$\delta_{uv} = r(u) - r(v), \quad g(\delta) = \mathbf{1} + \text{MLP}_g([\delta, |\delta|, \text{sign } \delta]), \quad (2)$$

$$h_v^{(t+1)} = \text{post}\left(\sum_{u \in N(v)} g(\delta_{uv}) \odot m_{uv}\right), \quad (3)$$

where $\odot$ is the Hadamard product, pre and post are MLPs (the term e_{uv} is omitted when there are no edge features), and MLP_g is a two-layer network $\text{Linear} \rightarrow \text{ReLU} \rightarrow \text{Linear}$ whose output has the channel dimension of m_{uv} . The output layer of MLP_g is initialised to zero, so $g \equiv \mathbf{1}$ is exactly realisable (and is the value at initialisation): with $g \equiv \mathbf{1}$ the layer is a plain summing message-passing layer.

The LDNA network. The wrapper (LDNA) applies a node encoder and, if present, an edge encoder; then a stack of L blocks, each a layer (1) to (3) followed by BatchNorm, a pointwise activation and dropout, with an optional per-layer residual $h_v^{(t+1)} += Wh_v^{(t)}$ (W the identity where the layer preserves width, a learnable linear map where it changes width); then a permutation-invariant readout in $\{\text{add}, \text{mean}, \text{max}, \text{attention}\}$ (add = sum, mean = average, max = coordinatewise maximum, attention = a softmax-weighted sum, PyG’s **AttentionalAggregation**) or the order-sensitive **gru** readout (a GRU over the sorted node sequence); and finally a prediction MLP. We treat the **gru** readout separately in Remark 5; the four listed readouts are all functions of the multiset $\{\{h_v^{(L)} : v \in V\}\}$.

Remark 1 (Inference-time normalisation). We analyse the network at inference. Dropout is then inactive, and BatchNorm is a fixed affine map $z \mapsto \gamma \odot (z - \mu)/\sigma + \beta$, which can be absorbed into the adjacent MLP; wherever a construction needs it we take this affine map to be the identity. We do not mention BatchNorm or dropout again.

Canonical rank. The rank r is supplied to Equation (2) in one of two modes. In both, only within-graph gaps $\delta_{uv} = r(u) - r(v)$ are ever used (all edges are intra-graph), so any per-graph additive offset in r is immaterial.

Feature mode. Nodes are first put into a fixed data-side canonical order by sorting each graph lexicographically by its raw feature rows (**sort_graph**); this fixes a total order $<_{\text{lex}}$ on the finite alphabet Σ . Let $X_G = \{x_v : v \in V\} \subseteq \Sigma$ be the *set* of distinct rows occurring in G and

$$\rho_G(x) = |\{x' \in X_G : x' <_{\text{lex}} x\}| \quad (4)$$

the tie-aware *dense* rank of a row (nodes sharing a row share a rank). The feature-mode rank is

$$r(v) = \rho_G(x_v) / n. \quad (5)$$

Normalisation is by $n = |V|$, the number of nodes, not the number of distinct rows.

Position mode. When the data already supplies a canonical total order of the nodes by construction (for instance the abstract-syntax-tree depth-first order of `ogbg-code2`), $r(v) = p(v)/n$, where $p(v) \in \{0, \dots, n-1\}$ is the position of v in that order.

The reference test. The 1-WL colour refinement with edge colours assigns

$$c_v^{(0)} = x_v, \quad c_v^{(t+1)} = \text{HASH}\left(c_v^{(t)}, \{\{(c_u^{(t)}, e_{uv}) : u \in N(v)\}\}\right), \quad (6)$$

with HASH injective; colours range over a finite set at each round. The round- $(t+1)$ partition refines the round- t partition, and it is monotone: two nodes with distinct round- t colours have distinct round- $(t+1)$ colours. We say G and H are *1-WL-indistinguishable*, $G \sim_{1\text{-WL}} H$, if for every t their colour histograms agree, $\{\{c_v^{(t)} : v \in V_G\}\} = \{\{c_w^{(t)} : w \in V_H\}\}$; this forces $|V_G| = |V_H|$. Because refinement is monotone, if the histograms differ at some round t^* they differ at every round $t \geq t^*$; in particular they differ at the stable round $T \leq N-1$ at which refinement terminates on all graphs of size $\leq N$.

Definition 1 (Distinguishing power). For a model class $\mathcal{M}$ (all parameters and widths), $\mathcal{M}$ *distinguishes* a pair (G, H) if there exist parameters θ with $\mathcal{M}_\theta(G) \neq \mathcal{M}_\theta(H)$. We say $\mathcal{M}$ is *at least as expressive as* 1-WL, $\mathcal{M} \succeq 1\text{-WL}$, if it distinguishes every 1-WL-distinguishable pair; $\mathcal{M}$ is *strictly more expressive* than 1-WL if in addition it distinguishes some pair with $G \sim_{1\text{-WL}} H$.

Two constructive lemmas. The proofs use only the following two elementary facts; neither uses universal approximation.

Lemma 1 (Exact finite interpolation). *Let $S \subset \mathbb{R}^d$ be finite and $\psi : S \rightarrow \mathbb{R}^m$ arbitrary. There is a two-layer ReLU network $F(z) = W_2 \sigma(W_1 z + b_1) + b_2$ ($\sigma = \text{ReLU}$) with $F|_S = \psi$.*

Proof. Choose $w \in \mathbb{R}^d$ with the scalars $t_s := \langle w, s \rangle$ pairwise distinct over $s \in S$; the excluded w form the finite union of hyperplanes $\{w : \langle w, s - s' \rangle = 0\}$, $s \neq s'$, so such w exist. Order $S = \{s_1, \dots, s_K\}$ so that $t_1 < \dots < t_K$, and write $y_k = \psi(s_k)$. It suffices to build, for each output coordinate j , a continuous piecewise-linear $f_j : \mathbb{R} \rightarrow \mathbb{R}$ with $f_j(t_k) = (y_k)_j$ and to set $F(z)_j = f_j(\langle w, z \rangle)$. Writing $a_k = ((y_{k+1})_j - (y_k)_j) / (t_{k+1} - t_k)$ for the slope on $[t_k, t_{k+1}]$ ($k = 1, \dots, K-1$) and setting $a_0 := 0$, define

$$f_j(t) = (y_1)_j + \sum_{k=1}^{K-1} (a_k - a_{k-1}) \sigma(t - t_k).$$

For $t \leq t_1$ every $\sigma(t - t_k)$ vanishes, so $f_j(t) = (y_1)_j$; in particular $f_j(t_1) = (y_1)_j$. On the interval (t_k, t_{k+1}) exactly the units $\sigma(t - t_1), \dots, \sigma(t - t_k)$ are active, so the slope of f_j telescopes to $\sum_{i=1}^k (a_i - a_{i-1}) = a_k$; hence there f_j is affine with slope a_k , and $f_j(t_{k+1}) = f_j(t_k) + a_k(t_{k+1} - t_k) = (y_{k+1})_j$, so by induction on k the map f_j interpolates the points $(t_k, (y_k)_j)$. Each of the $K - 1$ hidden units computes $\sigma(\langle w, z \rangle - t_k)$ (a single ReLU of an affine form), and the outer layer takes the stated linear combination with the constant $(y_1)_j$ in the output bias, so F is a two-layer ReLU network. $\square$

Lemma 2 (Sum-injective encodings). *Let A be finite, $|A| = K$, and fix a bijection $\iota : A \rightarrow \{1, \dots, K\}$. The one-hot map $\chi(a) = e_{\iota(a)} \in \mathbb{R}^K$ satisfies, for all finite multisets M, M' over A , $\sum_{a \in M} \chi(a) = \sum_{a \in M'} \chi(a) \iff M = M'$. Moreover, for multisets of size $\leq N$ the scalar $\sum_{a \in M} (N + 1)^{\iota(a)-1}$ is injective in M .*

Proof. $\sum_{a \in M} \chi(a)$ is the count vector $(M(a))_{a \in A}$, which determines M and conversely. For the scalar version, the multiplicities $M(a) \leq N < N + 1$ are the digits of a base- $(N + 1)$ integer, whose representation is unique. $\square$

The pre and post MLPs below are taken with at least one hidden layer (i.e. `num_pre_layers`, `num_post_layers` ≥ 2), a valid configuration; Lemma 1 is then applied to realise them exactly on their finite domains. The node and edge encoders are the implemented ones and are *not* assumed to be such MLPs: they need only be *injective* on the finite alphabet, which every implemented family provides — a single embedding table (choose distinct rows), an additive multi-column encoder (**AtomEncoder**, a sum of per-column embeddings placed on disjoint coordinate blocks, so the sum is a concatenation), or a scalar `Linear(1, d)` (any nonzero weight). Finally, in the implementation the pre and gate output width is tied to the layer’s input width, so “message width” below refers to the layer’s hidden width, which is free (all widths are quantified over in Definition 1).

1.2 A lower bound in every rank mode

Proposition 1 (Lower bound). *Under minimum degree one, with the **add** read-out, LDNA is at least as expressive as 1-WL, $LDNA \succeq 1\text{-WL}$, in every rank mode.*

Proof. Set the gate to the constant $g \equiv 1$ by zeroing the output layer of MLP_g (Equation (2)); the rank then plays no role and the argument is identical in every mode. The layer reduces to $h_v^{(t+1)} = \text{post}(\sum_{u \in N(v)} \text{pre}([h_v^{(t)}, h_u^{(t)}, e_{uv}]))$. Let C_t be the finite set of round- t 1-WL colours.

We claim that there are parameters and, for each t , a map $\alpha_t : C_t \rightarrow \mathbb{R}^{d_t}$ with $h_v^{(t)} = \alpha_t(c_v^{(t)})$ for every node of every graph of size $\leq N$, where α_0 is injective and, for $t \geq 1$, α_t is one-hot. (One layer already produces a one-hot code, and the readout argument below runs at some $L \geq 1$.)

Base. $c_v^{(0)} = x_v \in \Sigma$, and the node encoder gives $h_v^{(0)} = \alpha_0(x_v)$ with α_0 injective on the finite Σ (any of the implemented encoders, as above). Injectivity of α_0 is all that the step below uses at $t = 0$.

Step. Assume $h_v^{(t)} = \alpha_t(c_v^{(t)})$ with α_t injective (and one-hot when $t \geq 1$). Since α_t is invertible on its image, from $[h_v^{(t)}, h_u^{(t)}, e_{uv}]$ one can read off the triple $(c_v^{(t)}, c_u^{(t)}, e_{uv})$, an element of $C_t \times C_t \times \Xi$. Choose pre, via Lemma 1, to output the one-hot $\chi(c_v^{(t)}, c_u^{(t)}, e_{uv})$ of this triple. Because $c_v^{(t)}$ is constant over $u \in N(v)$, the aggregated vector

$$J_v = \sum_{u \in N(v)} \chi(c_v^{(t)}, c_u^{(t)}, e_{uv})$$

is supported on the block indexed by $c_v^{(t)}$, and within that block it is, by Lemma 2, the histogram of $\{(c_u^{(t)}, e_{uv}) : u \in N(v)\}$. Thus J_v encodes the pair $(c_v^{(t)}, \{(c_u^{(t)}, e_{uv})\})$ injectively; the aggregation is non-empty because no node is isolated. By Equation (6) and injectivity of HASH, this pair is in bijection with $c_v^{(t+1)}$, so J_v takes a distinct value for each colour $c_v^{(t+1)} \in C_{t+1}$, over a finite set of possible values. Choose post, via Lemma 1, to send each such J_v to the one-hot $\alpha_{t+1}(c_v^{(t+1)})$. One-hot codes are non-negative, so the intervening ReLU acts as the identity (Remark 1). Take the widths strictly increasing so that every residual projection is a learnable map, set to zero; the residual connections are thus present but inert here. This proves the claim.

Graph level. Run $L = \max(T, 1)$ layers, where $T \leq N - 1$ is the stable round; then $L \geq 1$, so $h_v^{(L)} = \alpha_L(c_v^{(L)})$ is one-hot, and the add readout returns $\sum_v \alpha_L(c_v^{(L)})$, the histogram of round- L colours, which by Lemma 2 is injective in the multiset $\{c_v^{(L)}\}$. If G, H are 1-WL-distinguishable they differ at some round, hence (by monotonicity) at round L , so the two readouts differ; an identity (or any injective) prediction head preserves the difference. Hence LDNA distinguishes (G, H) . $\square$

Remark 2 (Isolated nodes and the necessity of minimum degree one). The hypothesis $N(v) \neq \emptyset$ is essential to Proposition 1, not cosmetic, and isolated nodes do occur in practice (for instance single-atom fragments in molecular graphs). One might hope the residual could rescue them, but it cannot, because of the layer ordering in the implementation: the activation is applied to the convolution output *before* the residual is added, and for an isolated node the aggregation is $J_v \equiv 0$ for every choice of parameters. Its update therefore collapses to an affine recurrence $h_v^{(t+1)} = b_t + W_t h_v^{(t)}$, with $b_t = \text{ReLU}(\text{BN}(\text{post}(0)))$ a constant and W_t the residual projection, so $h_v^{(L)}$ is an *affine* function of $\text{enc}(x_v)$. Take moreover the node encoder to be affine in a scalar feature — a $\text{Linear}(1, d)$ encoder, a supported configuration — so $h_v^{(L)} = c + A x_v$ for a fixed vector c and matrix A . Then on any two graphs consisting only of isolated nodes the add readout returns $\sum_v (c + A x_v) = |V| c + A \sum_v x_v$, which depends only on the node count and the *sum* of features: the feature multisets $\{1, 3\}$ and $\{2, 2\}$

collide, though 1-WL separates them. Thus for the class with an affine-in-a-scalar encoder and the **add** readout, minimum degree one cannot be dropped from Proposition 1.

Remark 3 (Cardinality-blind readouts). The proposition uses **add**, which the class includes. The **mean** and **attention** readouts are normalised (softmax weights sum to one), so they see only the *distribution* of final embeddings, not their multiset with multiplicities. Consequently they cannot separate a graph G from the disjoint union $G \sqcup G$: the colour multiset of $G \sqcup G$ is that of G with every multiplicity doubled, so **mean** and **attention** return identical values on G and $G \sqcup G$, whereas 1-WL distinguishes them (their colour histograms, and indeed their sizes, differ). This is why the lower bound is stated for **add**; cf. the analogous discussion of the mean aggregator by Xu et al. [2019].

1.3 Exact characterisation via a rank-augmented test

Fix a rank function r . Define the *rank-augmented graph* G^r to have the same nodes, edges and features as G , with each directed edge $u \rightarrow v$ carrying the augmented colour (e_{uv}, δ_{uv}) , $\delta_{uv} = r(u) - r(v)$. Let 1-WL ^{r} denote edge-coloured 1-WL run on G^r : it is Equation (6) with the neighbour term replaced by $\{(c_u^{(t)}, e_{uv}, \delta_{uv}) : u \in N(v)\}$.

Proposition 2 (Characterisation). *For any fixed rank r , LDNA with rank r has exactly the distinguishing power of 1-WL ^{r} ; the $\succeq$ direction assumes minimum degree one.*

Proof. Upper bound ($\preceq$). Each LDNA message $g(\delta_{uv}) \odot m_{uv}$ depends only on $(h_v^{(t)}, h_u^{(t)}, e_{uv}, \delta_{uv})$, so a layer is a message-passing update on the edge-coloured graph G^r . By induction on t there is a map Φ_t (parameter-dependent, but the same function for all graphs) with $h_v^{(t)} = \Phi_t(c_v^{(t),r})$, where $c^{(\cdot),r}$ are the 1-WL ^{r} colours: the base is the encoder, and the step holds because Equations (1) to (3) turn $(c_v^{(t),r}, \{(c_u^{(t),r}, e_{uv}, \delta_{uv})\})$ into $h_v^{(t+1)}$, and that pair is determined by $c_v^{(t+1),r}$. Every permutation-invariant readout is a function of the final colour histogram, so graphs with equal 1-WL ^{r} histograms at all rounds receive equal outputs.

Lower bound ($\succeq$). Assuming minimum degree one, we adapt Proposition 1 to also encode δ . The gate is per-channel and depends on δ , while pre depends on (h_v, h_u, e) but not on δ ; their Hadamard product lets us form a *joint* one-hot of the quadruple $(c_v^{(t)}, c_u^{(t)}, e_{uv}, \delta_{uv})$. Concretely, δ ranges over a finite set D (there are finitely many graphs of size $\leq N$, hence finitely many gap values; for the model’s own rank modes $D \subseteq \{k/n : |k| < n \leq N\}$); index the message channels by (a, b, ε, j) with $(a, b, \varepsilon) \in C_t \times C_t \times \Xi$ and j indexing D . Let pre output, on channel (a, b, ε, j) , the indicator $\mathbf{1}[(c_v^{(t)}, c_u^{(t)}, e_{uv}) = (a, b, \varepsilon)]$ (independent of j), and let the gate output, on that channel, $\mathbf{1}[\delta_{uv} = \delta_j]$ (independent of (a, b, ε)): the map $\delta \mapsto [\delta, |\delta|, \text{sign } \delta]$ is injective on the finite D , so Lemma 1 realises this indicator (minus one) with MLP_g . The product on channel (a, b, ε, j) is

$\mathbf{1}[(c_v^{(t)}, c_u^{(t)}, e_{uv}, \delta_{uv}) = (a, b, \varepsilon, \delta_j)]$, a one-hot of the quadruple. Summing over $u \in N(v)$ (with $c_v^{(t)}$ constant) gives, exactly as in Proposition 1, an injective encoding of $(c_v^{(t)}, \{(c_u^{(t)}, e_{uv}, \delta_{uv})\})$, which by injectivity of HASH is in bijection with $c_v^{(t+1),r}$; post maps it to the one-hot $\alpha_{t+1}(c_v^{(t+1),r})$. Thus LDNA simulates 1-WL^r, and the add readout recovers its colour histogram, as before. $\square$

1.4 Feature mode collapses to 1-WL

Lemma 3 (Rank rigidity in feature mode). *If $G \sim_{1\text{-WL}} H$ then*

- (i) $n_G = n_H =: n$;
- (ii) *the multisets of raw feature rows of G and H coincide, hence the sets $X_G = X_H$ of distinct rows coincide and inherit the same order $<_{\text{lex}}$;*
- (iii) $\rho_G = \rho_H =: \rho$ as functions, and the feature-mode rank is $r(v) = \rho(x_v)/n$ in both graphs — one fixed function of a node’s own raw feature row, shared by G and H .

Proof. $G \sim_{1\text{-WL}} H$ requires equal round-0 histograms, $\{x_v : v \in V_G\} = \{x_w : w \in V_H\}$. Equal multiset cardinalities give (i); equal multisets have equal underlying sets $X_G = X_H$, and $<_{\text{lex}}$ is a fixed total order on Σ that restricts identically to the common set, giving (ii). By Equation (4), ρ_G depends only on $(X_G, <_{\text{lex}})$, so $\rho_G = \rho_H = \rho$; with the common n from (i), Equation (5) gives (iii). $\square$

Theorem 1 (Feature mode equals 1-WL). *In feature mode:*

- (a) *under minimum degree one, with the add readout, LDNA $\succeq$ 1-WL;*
- (b) *for every parameter setting and every permutation-invariant readout (add, mean, max, attention), $G \sim_{1\text{-WL}} H$ implies identical outputs on G and H .*

Hence feature-mode LDNA is exactly as expressive as 1-WL.

Proof. Part (a) is Proposition 1, which is rank-mode-agnostic. For part (b), fix $G \sim_{1\text{-WL}} H$ and any parameters. By Lemma 3 the gap

$$\delta_{uv} = \frac{\rho(x_u) - \rho(x_v)}{n} \quad (7)$$

is one fixed function of the raw endpoint rows, common to G and H .

We show by induction on t that there is a function ϕ_t (parameter-dependent, common to both graphs) with $h_v^{(t)} = \phi_t(c_v^{(t)})$ for all nodes of G and H . The base is $h_v^{(0)} = \text{encoder}(x_v)$ with $c_v^{(0)} = x_v$, so $\phi_0 = \text{encoder}$. For the step, recall that $c_v^{(t)}$ refines $c_v^{(0)} = x_v$, so $c_v^{(t)}$ determines x_v and hence $\rho(x_v)$; likewise for each neighbour. Therefore every ingredient of Equations (1) to (3) — the messages $m_{uv} = \text{pre}([\phi_t(c_v^{(t)}), \phi_t(c_u^{(t)}), e_{uv}])$, the gate $g(\delta_{uv})$ with δ_{uv} read off from

$(c_v^{(t)}, c_u^{(t)})$ via Equation (7), the sum over $u \in N(v)$, and post — is determined by $(c_v^{(t)}, \{(c_u^{(t)}, e_{uv}) : u \in N(v)\})$, which by injectivity of HASH is determined by $c_v^{(t+1)}$. The residual $W\phi_t(c_v^{(t)})$, the affine BatchNorm and the pointwise activation act node-wise and preserve this dependence (each is a function of $c_v^{(t+1)}$, which determines $c_v^{(t)}$). Set ϕ_{t+1} accordingly.

At the graph level, $G \sim_{1\text{-WL}} H$ gives equal final colour histograms, hence equal multisets $\{h_v^{(T)}\} = \{\phi_T(c_v^{(T)})\}$. The readouts **add**, **mean** (the sizes agree by Lemma 3(i)), **max** and **attention** are all functions of this multiset, so the graph embeddings agree; the prediction MLP is pointwise, so the outputs agree.

Equivalently, via Proposition 2: by Equation (7) the augmented edge colour (e_{uv}, δ_{uv}) is a function of the endpoints' round-0 colours, so 1-WL^r and 1-WL induce the same partition, and the characterisation collapses to plain 1-WL. Combining (a) and (b) proves the claim. $\square$

Corollary 1. *Consider attributed regular graphs — graphs whose nodes split into classes $V_1, \dots, V_k$, all nodes of class V_i sharing the attribute C_i , such that every node of class V_i has exactly r_{ij} neighbours in class V_j . Encode this as the configuration $\{(C_i, C_j, r_{ij}) : 1 \leq i, j \leq k\}$, which records the attributes as well as the counts. Then any two such graphs without edge features (or with edge features constant on each ordered pair of classes) that have equal configurations and equal class sizes are 1-WL-indistinguishable, and hence receive identical feature-mode LDNA outputs for every parameter setting.*

Proof. The partition into classes is equitable and each class carries a fixed attribute: the counts r_{ij} and the attributes C_i are constant across each class, and the edge colours (absent, or constant on each ordered class pair) add nothing. Hence 1-WL initialised at the attributes stabilises at this partition, and each round- t colour is a function of a node's class alone, determined by the configuration. Two such graphs with equal configurations and equal class sizes therefore have equal round-0 histograms and equal colour histograms at every round, i.e. $G \sim_{1\text{-WL}} H$. Apply Theorem 1(b). $\square$

1.5 Invariant tie-breaking does not help robustly

Theorem 1 shows that a rank read off the raw features is folded away. One might hope to strengthen the rank by sorting on a richer node quantity. The next proposition shows that any *isomorphism-invariant* choice of rank remains bounded by 1-WL, and fails outright on the classical hard instances.

Proposition 3 (Obstruction for invariant ranks). *(i) Let r be obtained by sorting nodes on any node invariant ι that is a function of the stable 1-WL colour κ (for instance the degree, or κ itself). Then LDNA with rank r is bounded by 1-WL.*

(ii) Let r be obtained by sorting on any isomorphism-invariant node function (possibly stronger than 1-WL, e.g. per-node subgraph counts). Then for

any pair of vertex-transitive graphs with $G \sim_{1\text{-WL}} H$, LDNA with rank r cannot distinguish G from H .

Proof. (i). Since ι is a function of κ , and the dense rank Equation (4) of ι -values divided by the common size is, for $G \sim_{1\text{-WL}} H$, a fixed function of κ (the multiset of κ , hence of ι , agrees across G and H), the gap $\delta_{uv} = f(\kappa(u), \kappa(v))$ is determined by the endpoints' stable colours. By Proposition 2, LDNA with r is bounded by 1-WL^r . We claim 1-WL^r and 1-WL induce the same partition. On one hand 1-WL^r refines 1-WL . On the other, the stable partition κ is equitable for G , and the added edge colours $\delta = f(\kappa(u), \kappa(v))$ are constant on each ordered pair of classes; hence for a node in class C the multiset $\{(\kappa(u), e_{uv}, \delta_{uv}) : u \in N(v)\}$ is a fixed function of the un-augmented multiset $\{(\kappa(u), e_{uv})\}$, which is constant across C . Thus κ is equitable for G^r as well, so 1-WL^r does not refine κ and equals it. To pass from equal partitions to equal graph-level histograms, check by induction on t that every 1-WL^r colour $c_v^{(t),r}$ is the image of $\kappa(v)$ under a map shared by G and H . Base: $c_v^{(0),r} = x_v$, and κ refines the raw feature, so x_v is a function of $\kappa(v)$. Step:

$$c_v^{(t+1),r} = \text{HASH}\left(c_v^{(t),r}, \{(\kappa(u), e_{uv}, \delta_{uv}) : u \in N(v)\}\right);$$

by the induction hypothesis and $\delta_{uv} = f(\kappa(u), \kappa(v))$ with f shared (the dense-rank map is shared, as above), the augmented neighbour multiset is a shared-map image of $\mu_v := \{(\kappa(u), e_{uv}) : u \in N(v)\}$, and μ_v is itself a fixed function of $\kappa(v)$ because κ is equitable; hence $c_v^{(t+1),r}$ is a shared function of $\kappa(v)$. Consequently each per-round 1-WL^r colour histogram is the image, under a shared map, of the corresponding κ -histogram. The κ -histograms agree across G and H (since $G \sim_{1\text{-WL}} H$ fixes every round histogram, including the stable one), so the 1-WL^r histograms agree too. Therefore LDNA with r distinguishes no pair that 1-WL cannot.

(ii). If ι is isomorphism-invariant and G is vertex-transitive, then for any u, v there is an automorphism π with $\pi(u) = v$, whence $\iota(u) = \iota(v)$; so ι is constant and all nodes are tied. Then r is constant, $\delta_{uv} \equiv 0$ on every edge, and the gate takes the single value $g(0) = \mathbf{1} + \text{MLP}_g([0, 0, 0])$, a constant that folds into post. LDNA thereby degenerates to the gate-free model of Proposition 1; equivalently $G^r = G$ (all $\delta = 0$), so by Proposition 2 LDNA with r is bounded by 1-WL on this pair and cannot distinguish $G \sim_{1\text{-WL}} H$. The classical 1-WL -hard pairs are vertex-transitive — for example the 4×4 rook's graph versus the Shrikhande graph, and the circular-skip-link (CSL) family — so invariant-based tie-breaking provably fails on them. $\square$

1.6 Canonical rank and strict separation

The obstruction in Proposition 3 is that an isomorphism-invariant rank must respect automorphisms and so ties every orbit. Breaking the symmetry requires a rank that is invariant across graphs but not across the automorphisms of a single graph — exactly what a canonical labelling provides.

Definition 2 (Injective canonical rank). A canonical labelling algorithm assigns to each attributed graph G a bijection $\pi_G : V \rightarrow \{0, \dots, n-1\}$ so that isomorphic graphs receive identical labelled graphs (a canonical form): for every isomorphism $f : G \rightarrow H$, relabelling G by π_G and H by π_H yields the same ordered attributed graph. The induced canonical rank is $r(v) = \pi_G(v)/n$; it is injective on the nodes of G .

Remark 4 (Properties of the canonical rank). Three points attach to Definition 2. *Well-definedness under automorphisms.* A canonizer may choose any representative labelling within an automorphism orbit; all such choices give the same canonical form, so the end-to-end model output is an exact, deterministic graph invariant. *Realisation.* The **position** rank mode implements this exactly when the data supply a canonical order by construction (the AST depth-first order of `ogbg-code2`). Alternatively the feature lexsort is completed into a full canonization by individualization–refinement [McKay and Piperno, 2014]: the lexsort is the initial colour classification, colour refinement is 1-WL, and what is added is a canonical individualization of the remaining ties. *Complexity.* Graph canonization has no known polynomial worst-case algorithm — the best bound is quasi-polynomial [Babai, 2016] — but is fast in practice on benchmark graph classes [McKay and Piperno, 2014]; it is a one-time, data-side preprocessing step, in the same pipeline position as the existing lexsort.

Theorem 2 (Strictness under canonical rank). *With an injective canonical rank, under minimum degree one, and the **add** readout, LDNA is strictly more expressive than 1-WL.*

Proof. That LDNA $\succeq$ 1-WL is Proposition 1, which holds in every rank mode. It remains to exhibit a 1-WL-indistinguishable pair that LDNA separates.

Let $G_1 = C_6$ be the 6-cycle and $G_2 = C_3 \sqcup C_3$ two disjoint triangles, all node features equal to a single symbol and no edge features. Both are 2-regular on six nodes (so minimum degree one holds) with identical initial colours. By induction, at every 1-WL round all six nodes of either graph share one colour: at round 0 this is the single symbol, and if all nodes share colour a at round t , then each node’s neighbour multiset is $\{\{a, a\}\}$, so HASH returns one common colour at round $t+1$. The round- t histograms are therefore $\{\{a_t^{(6)}\}\}$ for both graphs and agree at every t ; hence $G_1 \sim_{1\text{-WL}} G_2$.

A canonizer is free to fix one representative labelling per isomorphism class. Choose $\pi(C_6)$ to be the cyclic order (positions 0, 1, 2, 3, 4, 5 along the cycle) and $\pi(C_3 \sqcup C_3)$ to place the two triangles on $\{0, 1, 2\}$ and $\{3, 4, 5\}$. With $r(v) = \pi(v)/6$, the normalised gaps $|\delta|$ over the undirected edges are

$$\begin{aligned}
C_6 : \quad & \underbrace{\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}}_{\text{edges } (0,1), \dots, (4,5)}, \underbrace{\frac{5}{6}}_{(5,0)} \quad \left(\sum = \frac{10}{6} \right); \\
C_3 \sqcup C_3 : \quad & \underbrace{\frac{1}{6}, \frac{1}{6}}_{(0,1), (1,2)}, \underbrace{\frac{2}{6}}_{(0,2)}, \underbrace{\frac{1}{6}, \frac{1}{6}}_{(3,4), (4,5)}, \underbrace{\frac{2}{6}}_{(3,5)} \quad \left(\sum = \frac{8}{6} \right).
\end{aligned}$$

Now fix parameters for a single layer on one channel. Take the node encoder to output the constant 0; then $h_v^{(0)} = 0$ and the residual, if present, contributes 0 (it is the identity on equal widths). Let $\text{pre} \equiv 1$ (a constant map: zero the output-layer weight and set its bias to one), so $m_{uv} = 1$. Realise the gate $g(\delta) = 1 + |\delta|$: the first layer of MLP_g selects the $|\delta|$ coordinate of $[\delta, |\delta|, \text{sign } \delta]$, the ReLU leaves it unchanged since $|\delta| \geq 0$, and the second layer copies it with weight one. Let post be the two-layer MLP computing the identity on the non-negative achievable values (both layers weight one; the intervening ReLU acts as the identity because the values are non-negative). Every intermediate value is non-negative, so the block activation acts as the identity too (Remark 1). Each node then has

$$h'_v = \sum_{u \in N(v)} g(\delta_{uv}) m_{uv} = \sum_{u \in N(v)} (1 + |\delta_{uv}|),$$

and, writing m for the number of *undirected* edges (so E contains $2m$ directed edges), the **add** readout returns, summing over directed edges (each undirected edge in both orientations, with $|\delta_{uv}| = |\delta_{vu}|$),

$$\text{out}(G) = \sum_v \sum_{u \in N(v)} (1 + |\delta_{uv}|) = 2m + 2 \sum_{\text{undirected } uv} |\delta_{uv}|.$$

Both graphs have $m = 6$ undirected edges, so

$$\text{out}(C_6) = 12 + \frac{20}{6} = \frac{92}{6}, \quad \text{out}(C_3 \sqcup C_3) = 12 + \frac{16}{6} = \frac{88}{6},$$

which differ. LDNA with this canonical rank therefore distinguishes the 1-WL-indistinguishable pair $(C_6, C_3 \sqcup C_3)$, and by Definition 1 it is strictly more expressive than 1-WL. $\square$

The theorem asserts the *existence* of a canonical labelling under which LDNA separates the pair; fixing representatives is legitimate precisely because a canonizer is a choice of one representative per isomorphism class.

1.7 Closing remarks

Remark 5 (The **gru** readout). The **gru** readout runs a recurrent network over the sorted node sequence and is order-sensitive. Under the feature sort, the order within a block of tied (identical-row) nodes is inherited from the arbitrary input labelling, so on graphs with ties the resulting graph function is not permutation-invariant and is excluded from the statements above. On tie-free graphs (all rows distinct) there is no expressiveness to be gained either: with an injective initial colouring, 1-WL is complete after one round. Indeed, the round-1 colour of a node is $(x_v, \{\{x_u : u \in N(v)\}\})$; since the features are distinct identifiers, matching two graphs' nodes by feature is a bijection, and equal round-1 colour multisets force equal neighbour-feature sets at every node, i.e. equal labelled adjacency, hence isomorphism. So any two non-isomorphic tie-free graphs are already separated by 1-WL.

Remark 6 (A trilemma). Informally, and as a synthesis of the three results above rather than a separate theorem: of the three properties — deterministic exact invariance, individualization beyond 1-WL, and cheap worst-case preprocessing — a rank scheme can secure at most two. Arbitrary or random tie-breaking (random node individualization) attains the extra expressiveness but is invariant only in distribution [Abboud et al., 2021]; invariant tie-breaks keep exact invariance but fail on vertex-transitive pairs (Proposition 3); canonical individualization keeps both invariance and expressiveness at the cost of a canonization step (Definition 2).

Remark 7 (Capacity, not learnability). All results above concern representational capacity — the existence of parameters realising a given behaviour. They say nothing about whether such parameters are found by optimization, nor about generalization.

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